

GAITHERSBURG MONTGOMERY, MD, USA

WMO: 720334

Lat: 39.167N

Lon: 77.167W

Elev: 164

StdP: 99.37

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 93764

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.1	-20.2	0.6	-10.9	-17.5	0.8	-8.0	11.3	0.8	10.3	-0.6	2.7	300	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.9	24.2	32.2	23.8	31.0	23.3	25.4	31.6	24.7	30.3	24.0	28.8	2.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.7	28.5	22.9	17.9	27.4	22.4	17.5	26.9	78.5	31.3	75.4	30.6	72.6	28.8	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.5	6.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-15.8	26.8	2.1	1.0	-17.3	27.5	-18.5	28.1	-19.7	28.6	-21.2	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	0.0	2.0	6.2	12.4	18.7	22.4	24.6	23.4	20.1	13.8	7.1	3.3		
	DBStd	9.66	5.48	5.75	5.54	4.92	4.38	3.15	2.77	2.57	3.74	4.64	4.58	4.94		
	HDD10.0	1062	313	233	146	28	1	0	0	0	16	109	216			
	HDD18.3	2634	568	458	377	187	53	6	0	1	23	157	337	466		
	CDD10.0	2115	3	8	28	101	269	372	452	414	304	132	23	9		
	CDD18.3	645	0	0	1	10	63	128	194	156	78	15	1	0		
	CDH23.3	5555	0	2	10	138	536	1109	1803	1264	588	103	1	0		
	CDH26.7	1922	0	0	0	37	151	366	724	429	194	22	0	0		
(14) Wind	WSAvg	2.2	2.8	2.9	2.9	2.7	2.1	1.8	1.6	1.5	1.5	2.0	2.3	2.5		
(15) Precipitation	PrecAvg	1137	86	67	97	91	110	108	104	102	101	98	77	92		
	PrecMax	1682	157	153	202	159	224	272	200	205	275	244	171	168		
	PrecMin	850	44	10	0	49	31	14	30	24	0	0	22	44		
	PrecStd	185	30	35	42	31	52	62	42	43	62	59	38	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	22.1	24.6	31.0	32.3	33.8	36.0	33.5	32.9	28.9	22.3	19.1		
		MCWB	13.2	15.2	15.4	18.1	23.4	24.6	25.4	24.5	24.0	22.5	16.0	16.4		
	2%	DB	13.9	17.7	22.0	27.4	30.3	32.2	33.4	32.3	31.2	26.3	19.1	17.1		
		MCWB	11.0	12.2	14.7	17.4	22.1	23.8	24.5	24.1	23.0	20.5	14.5	14.7		
	5%	DB	11.2	13.9	18.6	24.0	27.9	30.9	32.3	30.9	28.8	23.7	17.3	13.6		
		MCWB	8.0	9.3	12.7	15.9	20.7	23.2	24.1	23.5	22.5	18.8	12.7	10.8		
	10%	DB	7.8	11.0	16.0	22.0	26.6	28.8	31.0	29.0	27.1	21.8	14.8	11.0		
		MCWB	5.1	7.0	10.2	15.0	19.8	22.2	23.7	22.5	20.7	17.9	10.9	8.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	16.4	17.0	20.2	24.2	25.4	26.8	25.9	25.3	23.5	17.7	17.1		
		MCDB	16.9	20.2	22.3	25.3	30.3	32.1	33.6	31.8	30.9	27.2	20.8	18.2		
	2%	WB	11.7	13.3	15.3	18.6	23.1	24.5	25.7	25.0	24.1	21.8	15.4	14.9		
		MCDB	14.0	16.9	20.1	25.3	28.4	30.7	32.1	30.6	28.9	24.6	18.1	17.1		
	5%	WB	8.0	9.9	13.3	17.0	21.9	23.7	25.0	24.1	23.2	20.3	13.6	11.7		
		MCDB	10.1	13.7	18.3	23.3	26.9	29.0	30.7	28.6	27.3	23.0	16.4	13.4		
	10%	WB	5.4	7.2	10.9	15.4	20.7	22.9	24.1	23.4	22.1	18.2	11.8	8.3		
		MCDB	7.9	10.4	14.9	20.7	25.0	27.4	28.9	27.5	25.2	20.8	13.9	9.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.3	10.4	11.9	11.0	10.8	10.8	10.5	10.3	10.6	10.0	8.2		
		MCDBR	12.6	14.0	15.3	15.4	12.9	12.5	12.2	12.0	12.5	12.9	12.9	11.0		
	5% WB	MCWBR	10.3	10.2	9.6	8.1	6.3	5.6	4.9	5.1	5.8	7.3	8.6	9.5		
		MCWBR	11.6	12.5	13.5	13.5	10.9	11.1	10.7	10.1	10.6	10.0	10.4	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.318	0.352	0.399	0.440	0.499	0.516	0.514	0.431	0.358	0.319	0.305		
	taud		2.472	2.412	2.390	2.276	2.191	2.064	2.036	2.023	2.232	2.441	2.510	2.504		
	Ebn at Noon		872	904	905	877	844	792	774	763	810	847	847	843		
	Edh at Noon		79	96	109	130	145	165	168	166	127	93	76	72		
(44) All-Sky Solar Radiation	RadAvg		2.02	2.88	3.82	4.86	5.45	6.04	5.95	5.29	4.32	3.21	2.31	1.77		
	RadStd		0.17	0.33	0.26	0.38	0.61	0.46	0.32	0.28	0.53	0.36	0.20	0.18		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (72 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	+117	+66

Nomenclature: See separate page